QUIRINO HIGH SCHOOL

TECHPRO • GRADE 11 ACTIVITY SHEET

CSS COMBINATORS, PSEUDO-CLASSES & PSEUDO-ELEMENTS

Selector relationships • Interactive states • Structural selectors • Generated content

Name: Ramo, Jeric Y. Date: 10 / 15 / 26
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Learning Objectives

- Explain how CSS combinators select elements according to relationships in HTML.
- Apply interactive and structural pseudo-classes to links, buttons, inputs, and list items.
- Use pseudo-elements to style the first letter or first line and add generated content.
- Combine selectors to create a clear, interactive TechPro web page.

Before You Begin

Use Visual Studio Code and a browser. Create a folder named CSS_Selector_Lab, save the starter code as index.html, and refresh the browser after each change.

Starter Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TechPro Selector Lab</title><style>
body{font-family:Arial,sans-serif;margin:0;line-height:1.6;color:#222}
header{background:#6b1024;color:white;padding:20px;text-align:center}main{padding:20px}
.notice{padding:12px;background:#f8e9ed;border-left:4px solid #6b1024}
.card{border:1px solid #bbb;padding:12px;margin:12px 0}.menu a{display:inline-block;padding:8px 12px;color:#6b1024}
.task-list li{margin:6px 0}input{padding:8px;margin:5px 0}
</style></head><body>
<header><h1>TechPro Selector Lab</h1><p>Practice combinators, pseudo-classes, and pseudo-elements.</p></header><main>
<nav class="menu"><a href="#combinators">Combinators</a><a href="#states">Interactive States</a><a href="#structure">Structure</a><a href="#content">Pseudo-elements</a></nav>
<section id="combinators"><h2>Combinator Practice</h2><div class="card"><p>Paragraph directly inside the card.</p><div><p>Paragraph nested in another div.</p></div></div><p>Paragraph after the card.</p></section>
<section id="states"><h2>Interactive States</h2><p><a href="#structure">Try this link</a></p><button type="button">Move pointer or click</button><p><label for="student-name">Student name:</label></p><input id="student-name" type="text" placeholder="Click here and type"></section>
<section id="structure"><h2>Structural Selectors</h2><ul class="task-list"><li>Open Visual Studio Code</li><li>Create index.html</li><li>Write the HTML structure</li><li>Add CSS rules</li><li>Save and refresh</li></ul></section>
<section id="content"><h2>Generated and Text Content</h2><p class="intro">Technology skills grow through practice. Test each selector and observe the changes.</p><p class="tip">Remember: use pseudo-classes for states and pseudo-elements to style part of an element or generate content.</p></section>
</main></body></html>
```

PART A — CSS Combinators

1. Add .card p { color: #6b1024; } and observe that paragraphs inside the card are selected, including the nested paragraph.
2. Add .card > p { font-weight: bold; } and compare direct children with nested descendants.
3. Add #combinators h2 + .card { background: #f8e9ed; } to style the card immediately after the heading.
4. Add .card ~ p { border: 1px dashed #6b1024; } and identify which paragraph is selected.
5. Explain the four combinators: space (descendant), > (child), + (next sibling), and ~ (subsequent sibling).

PART B — Interactive Pseudo-classes

6. Add button:hover { background: #6b1024; color: white; cursor: pointer; } and move the pointer over the button.
7. Add button:active { transform: scale(0.98); } and click the button.
8. Add input:focus { outline: 3px solid #d6a0ad; background: #fff8fa; } then click the input and type.
9. Style links using :link, :visited, :hover, and :active. Keep :hover after :link and :visited in your CSS, then test the states.
10. Explain how :hover, :focus, :active, :link, and :visited differ.

PART C — Structural Pseudo-classes

11. Add .task-list li:first-child { color: #6b1024; font-weight: bold; } to style the first item.
12. Add .task-list li:last-child { text-decoration: underline; } to style the last item.
13. Add .task-list li:nth-child(even) { background: #f8e9ed; } to style alternating items.
14. Try .task-list li:nth-child(3) { border-left: 4px solid #6b1024; padding-left: 8px; } and identify the selected item.
15. Explain how :first-child, :last-child, and :nth-child() select elements.

PART D — Pseudo-elements for Text

16. Add .intro::first-letter { color: #6b1024; font-size: 220%; font-weight: bold; float: left; line-height: 1; padding-right: 4px; }
17. Add .tip::first-line { font-variant: small-caps; color: #6b1024; } Resize the browser and observe the first line.
18. Remember: ::first-letter and ::first-line apply to block-level elements.

PART E — Pseudo-elements for Content

19. Add #content h2::before { content: '★ '; color: #b8860b; } to add a star before the heading.
20. Add #content h2::after { content: ' ✓'; color: #6b1024; } to add a check mark after the heading.
21. Add .task-list li::marker { color: #6b1024; font-size: 1.2em; } to style list markers.
22. Add ::selection { color: white; background: #6b1024; } and select text with your mouse.
23. Use generated content for decoration; keep essential information in the HTML itself.

PART F — Selector Combination Challenge

24. Style the first paragraph directly inside .card using a child combinator and a structural pseudo-class.
25. Make menu links change background on hover and show a visible focus outline for keyboard users.
26. Use ::before or ::after to decorate a heading without editing its HTML text.
27. Use ::first-letter or ::first-line to style the intro paragraph, then test the page by interacting with the links, button, input, and list.

MINI-PROJECT — TechPro Student Resource Hub

Build a student resource page that demonstrates the selector techniques practiced.

- Create a TechPro Student Resource Hub with a header, navigation links, task list, and information cards.
- Use all four combinators at least once: space, >, +, and ~.
- Use at least three pseudo-classes, including :hover and :focus.
- Use two structural pseudo-classes, such as :first-child, :last-child, or :nth-child().
- Use ::first-letter or ::first-line and add decoration with ::before or ::after.
- Style list markers and selected text with ::marker and ::selection.

Check Your Work

☐ My page opens correctly in a browser.

☐ I used all four CSS combinators.

☐ I tested hover, focus, and active states.

☐ I styled list items with structural pseudo-classes.

☐ I used text and generated-content pseudo-elements.

☐ I styled markers and selected text.

☐ Links and form controls have visible interaction states.

Review Questions

1. What relationship does the descendant combinator (space) select?

   selects all elements that are descendants of a specified element

2. How is the child combinator (>) different from the descendant combinator?

child combinator > selects only direct children of a specified element, whereas the descendant combinator selects any descendant regardless of nesting depth

3. What does the next-sibling combinator (+) select?

   selects an element that is the direct next sibling of a specified element

4. What does the subsequent-sibling combinator (~) select?

   selects all sibling elements that follow a specified element

5. What is a CSS pseudo-class?

   keyword added to a selector that specifies a special state of the targeted element

6. How do :hover and :focus differ?

:hover applies when the user places the mouse pointer over an element, while :focus applies when an element receives focus like when an input field is clicked or selected using keyboard navigation)

7. What does :nth-child(even) select?

   selects every child element that is at an even-numbered index position

8. What is the difference between a pseudo-class and a pseudo-element?

pseudo-class targets an existing element in a specific state or structural position using a single colon, while a pseudo-element styles a specific part of an element or inserts generated content using double colons

9. What do ::first-letter and ::first-line style?

:first-letter styles the very first letter of a block-level element, and :first-line styles its first line of text

10. How do ::before and ::after use the content property?

they use the content property to insert generated content like text, symbols, or decorative elements) right before or after an element's actual content

Quick Review

- Combinators: space = descendant; > = direct child; + = immediate next sibling; ~ = following siblings.
- Pseudo-classes use one colon (:) and target a state or structural position.

- Interactive examples: :hover, :focus, :active, :link, :visited.
- Structural examples: :first-child, :last-child, :nth-child().
- Pseudo-elements use double colons (::) to style part of an element or add generated content.
- Text pseudo-elements: ::first-letter and ::first-line.
- Content pseudo-elements: ::before, ::after, ::marker, and ::selection.

References

- https://www.w3schools.com/css/css_combinators.asp
- https://www.w3schools.com/css/css_pseudo_classes.asp
- https://www.w3schools.com/css/css_pseudo_classes_interactive.asp
- https://www.w3schools.com/css/css_pseudo_classes_structural.asp
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